

BEng (Hons) Control and Automation

Automation and Robotics

CRN: 50295

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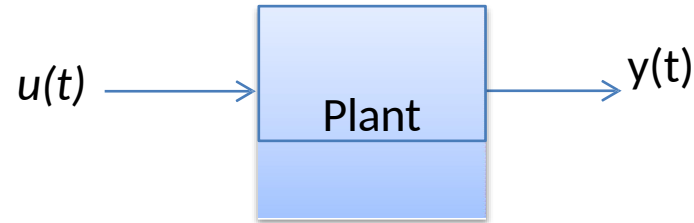
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Lecture Content:

Transfer Functions

Transfer Functions

The Transfer Function is the ratio of Laplace transform of the output to the Laplace transform of the input. Considering all initial conditions to zero.



If $\star u(t) = U(S)$
and

$\star y(t) = Y(S)$

- Where $\star$ is the Laplace operator.
- Then the transfer function $G(S)$ of the plant is given as

$$G(S) = \frac{Y(S)}{U(S)}$$



Laplace Transforms

By the use of Laplace transforms we can convert many common functions into an algebraic function of the complex variable s .

- For example

$$\star \sin \omega t = \frac{\omega}{s^2 + \omega^2}$$

Or

$$\star e^{-at} = \frac{1}{s + a}$$

- Where s is a complex variable (complex frequency) and is given as

$$s = \sigma + j\phi$$

Laplace Transform of Derivatives

Not only common function can be converted into simple algebraic expressions but calculus operations can also be converted into algebraic expressions.

- For example

$$\star \frac{dx(t)}{dt} = sX(S) - x(0)$$

$$\star \frac{d^2}{dt^2} x(t) = s^2 X(S) - x(0) - \frac{dx(0)}{dt}$$

- In general

$$\star \frac{d^n}{dt^n} x(t) = s^n X(S) - s^{n-1} x(0) - \dots - x^{n-1}(0)$$

- Where $x(0)$ is the initial condition of the system.

Laplace Transform of Integrals

The time domain integral becomes division by **s** in frequency domain.

$$\downarrow \\ X(s) = \frac{1}{s} \int_0^\infty x(t) dt$$

Calculation of the Transfer Function

Consider the following ODE where $y(t)$ is input of the system and $x(t)$ is the output.

$$A \frac{d^2 x(t)}{dt^2} = C \frac{dy(t)}{dt} - B \frac{dx(t)}{dt}$$

- or

$$Ax''(t) = Cy'(t) - Bx'(t)$$

- Taking the Laplace transform on either sides

$$A[s^2 X(s) - sx(0) - x'(0)] = C[sY(s) - y(0)] - B[sX(s) - x(0)]$$

Calculation of the Transfer Function

Considering Initial conditions to zero in order to find the transfer function of the system

$$A[s^2 X(s) - sx(0) - x'(0)] = C[sY(s) - y(0)] - B[sX(s) - x(0)]$$

Becomes

$$As^2 X(s) = \quad - \quad CsY(s) - BsX(s)$$

Rearranging the above equation

$$As^2 X(s) + BsX(s) = CsY(s)$$

$$X(s) [As^2 + Bs] = CsY(s)$$

$$\frac{X(s)}{Y(s)} = \frac{Cs}{As^2 + Bs} = \frac{C}{As + B}$$

$$B$$

Transfer Functions

In general

$$\begin{aligned} a_0 y^{(n)} + a_1 y^{(n-1)} + \cdots + a_{n-1} \dot{y} + a_n y \\ = b_0 x^{(m)} + b_1 x^{(m-1)} + \cdots + b_{m-1} \dot{x} + b_m x \quad (n \geq m) \end{aligned}$$

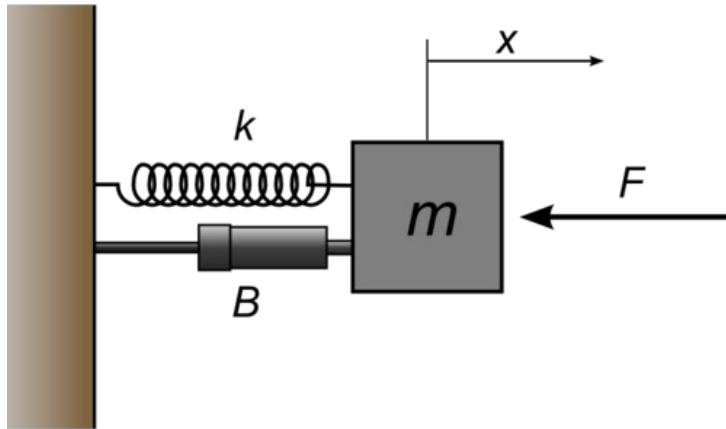
- Where x is the input of the system and y is the output of the system.

$$\begin{aligned} \text{Transfer function} = G(s) &= \frac{\mathcal{L}[\text{output}]}{\mathcal{L}[\text{input}]} \bigg|_{\text{zero initial conditions}} \\ &= \frac{Y(s)}{X(s)} = \frac{b_0 s^m + b_1 s^{m-1} + \cdots + b_{m-1} s + b_m}{a_0 s^n + a_1 s^{n-1} + \cdots + a_{n-1} s + a_n} \end{aligned}$$

- When the order of the denominator polynomial is greater than the numerator polynomial the transfer function is said to be 'proper'.

A Basic Mechanical System

Consider the following basic mechanical system



i)

**Translational
Spring**



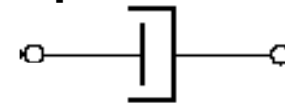
ii)

Translational Mass



iii)

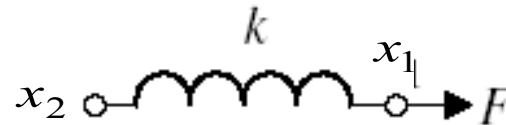
**Translational
Damper**



A Translational Spring

A translational spring is a mechanical element that can be deformed by an external force such that the deformation is directly proportional to the force applied to it.

- If F is the applied force



- Then x_1 is the deformation if $x_2 = 0$

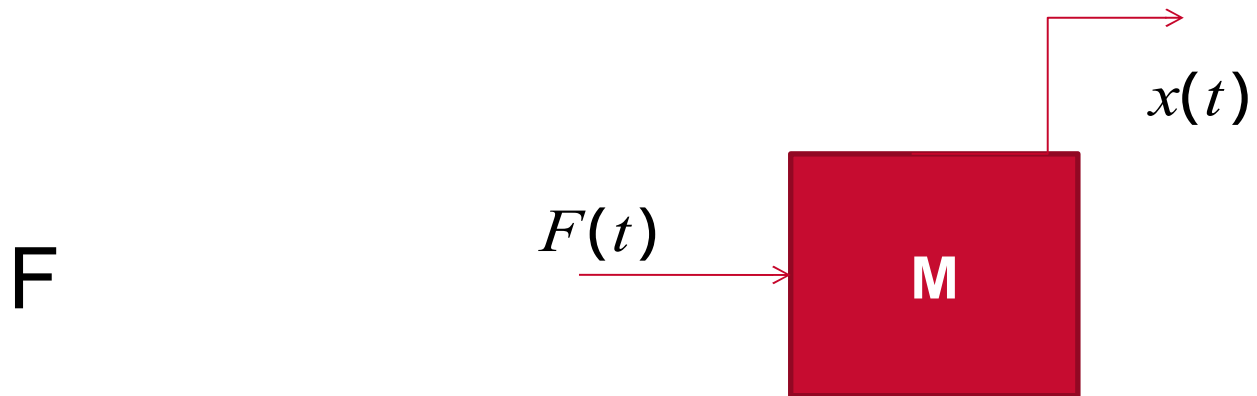


- Or $(x_1 - x_2)$ is the deformation.
- The equation of motion is given as $F = k(x_1 - x_2)$
- Where k is stiffness of spring expressed in N/m

A Translational Mass

A translational Mass is an inertia element.

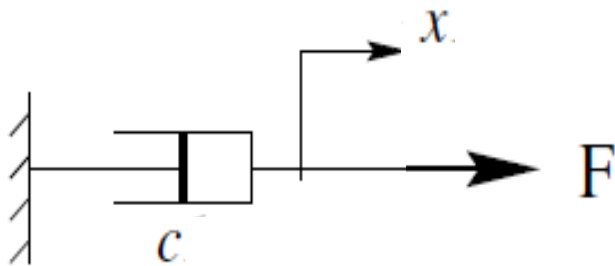
- If a force F is applied to a mass and it is displaced to x meters then the relation b/w force and displacements is given by Newton's law.



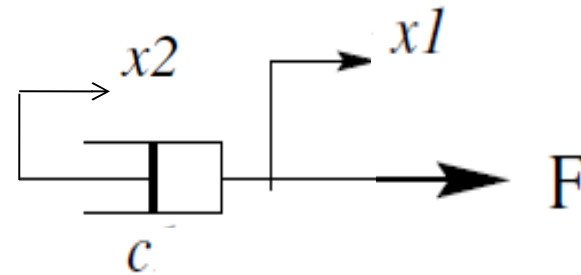
A Translational Damper

The damper opposes the rate of change of motion

- All the materials exhibit the property of damping to some extent.



F

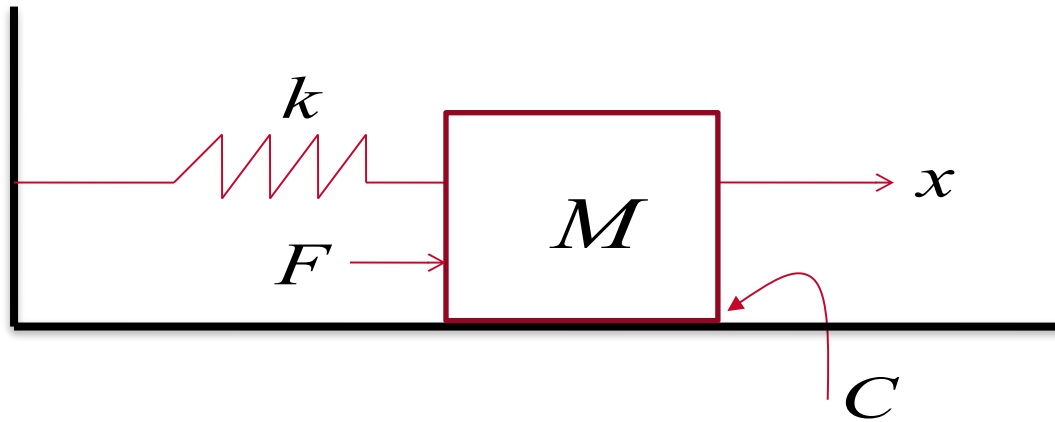


F -)

- Where C is damping coefficient (N/ms^{-1}).

Example

Consider the following system



Using a Free Body diagram



$$F = f_k + f_M + f_C$$

Example

- Differential equation of the system is:

$$F + C + k$$

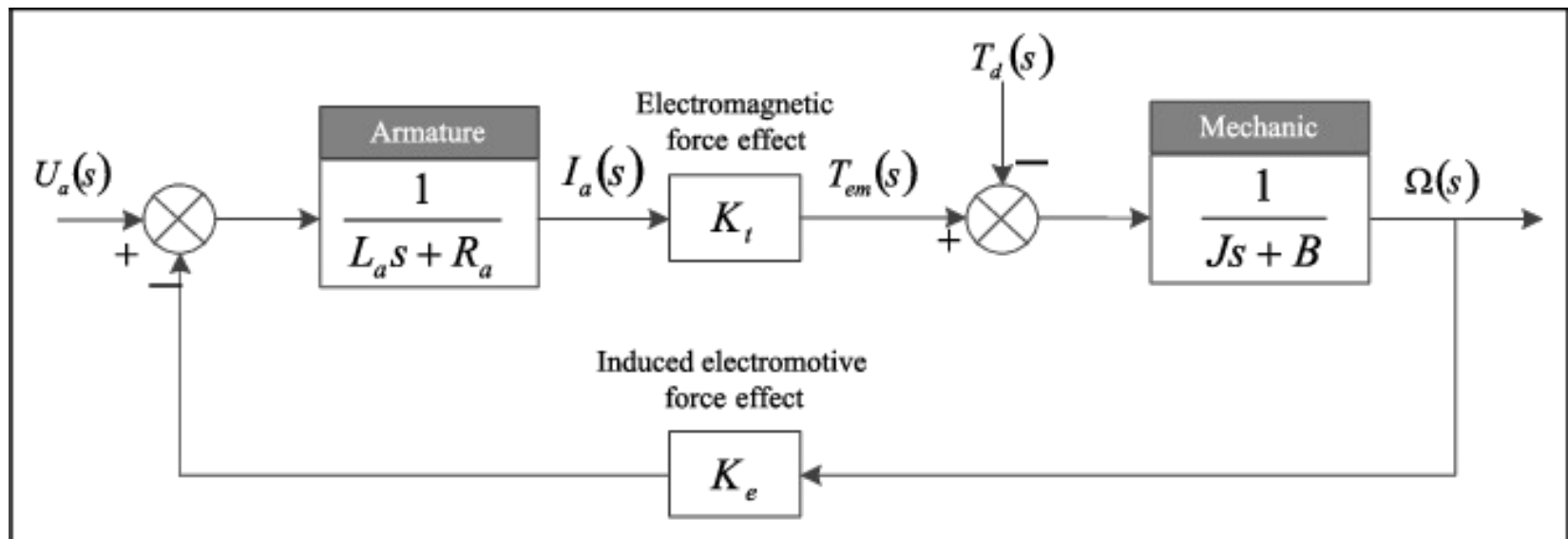
Taking the Laplace Transform of both sides and ignoring Initial conditions (set to zero) we get:

$$F(s) = Ms^2 X(s) + CsX(s) + kX(s)$$

$$\frac{X(s)}{F(s)} = \frac{1}{Ms^2 + Cs + k}$$

Application

- An example of how transfer functions are applied to dc motors



Poles and Zeros

$$\frac{X(s)}{F(s)} = \frac{1}{Ms^2 + Cs + k}$$

- Roots of denominator polynomial of a transfer function are called 'poles'.
- And the roots of numerator polynomials of a transfer function are called 'zeros'.
- Poles of the system are represented by 'x' and zeros of the system are represented by 'o'.
- System order is always equal to number of poles of the transfer function.
- Poles are also defined as "it is the frequency at which the system becomes infinite". Hence the name pole where the field is infinite.
- And zeros are the frequency at which system becomes 0.

Example

$$\frac{X(s)}{F(s)} = \frac{1}{Ms^2 + Cs + k}$$

- if

$$M = 1000 \text{ kg}$$

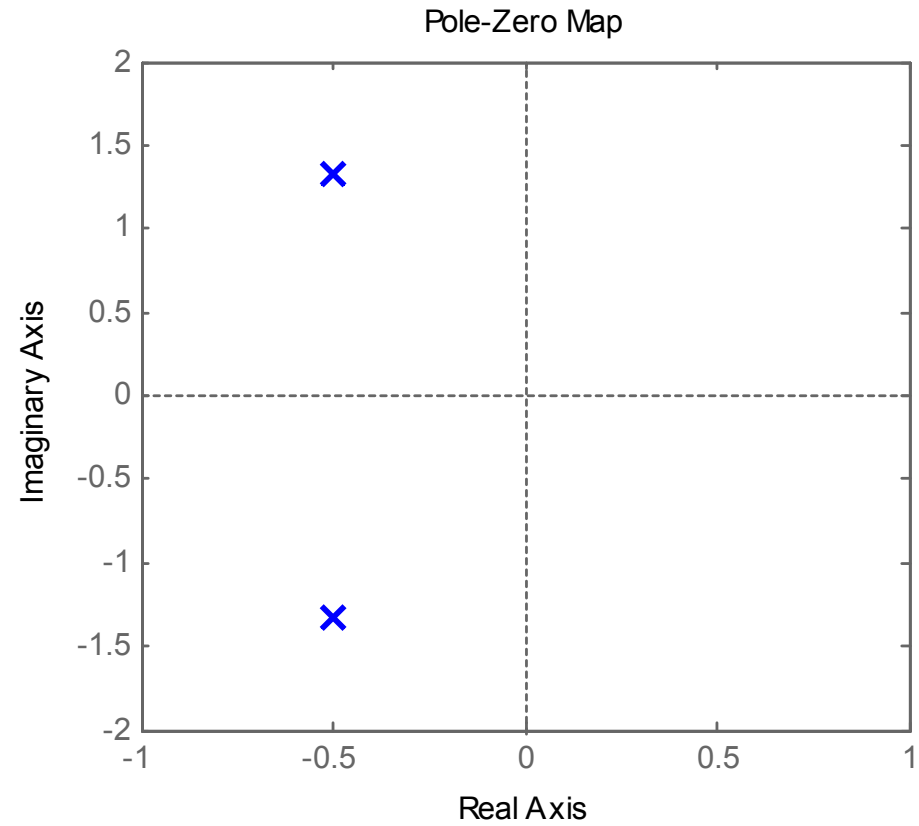
$$k = 2000 \text{ Nm}^{-1}$$

$$C = 1000 \text{ N / ms}^{-1}$$

Divide through by 1000

$$\frac{X(s)}{F(s)} = \frac{0.001}{s^2 + s + 2}$$

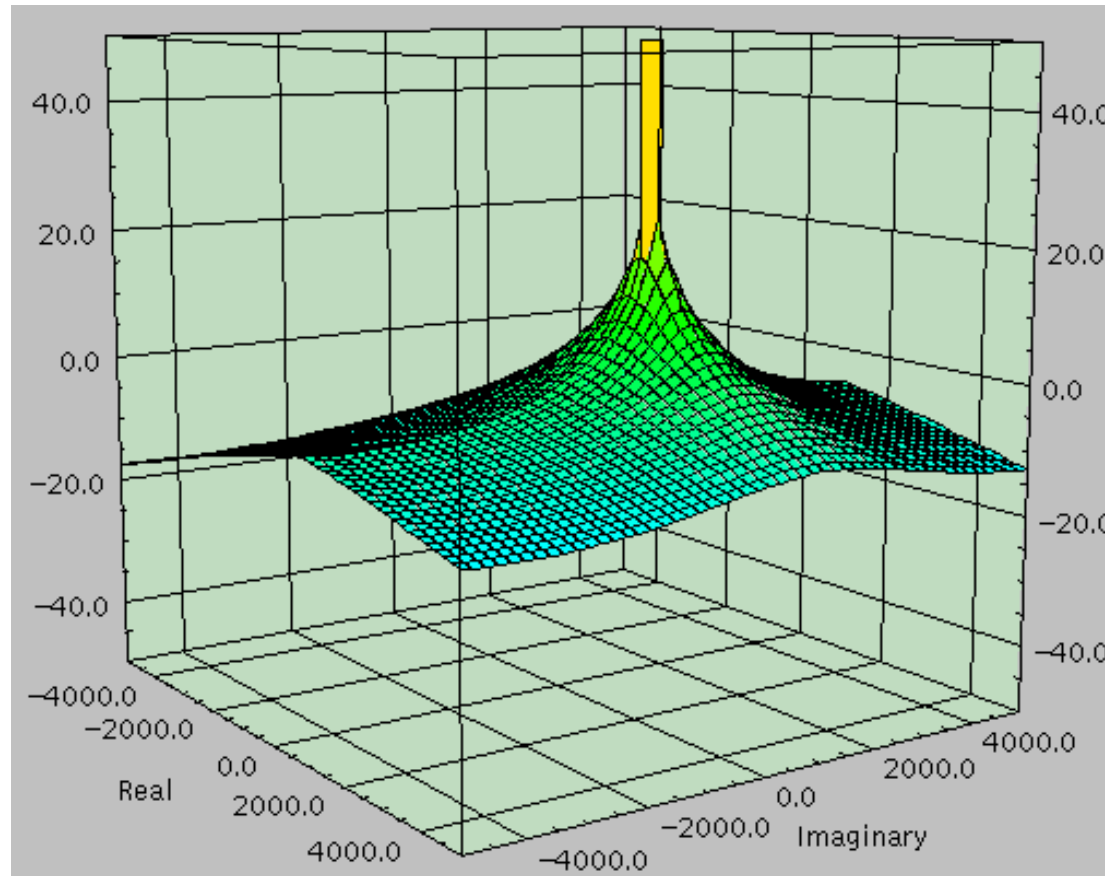
Apply $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ and find roots



Poles and Zeros

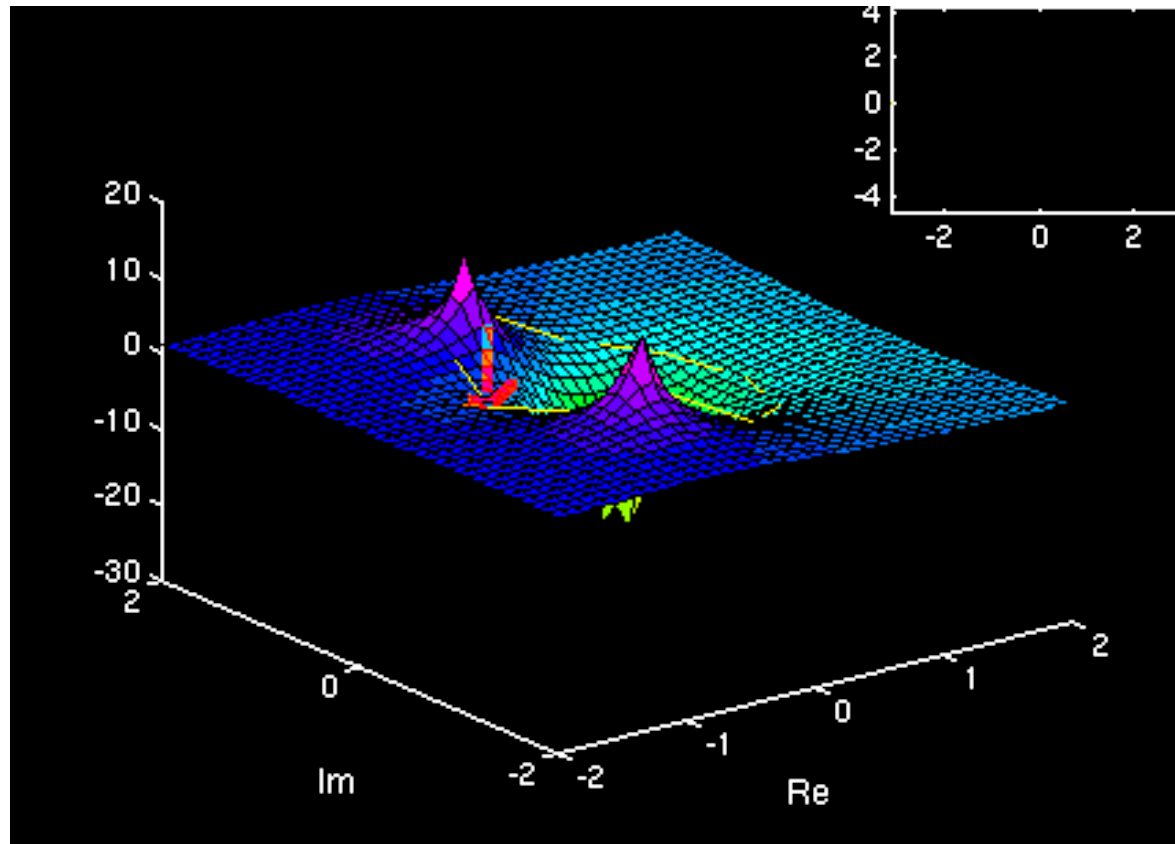
The relationship between poles and zeros and the frequency response of a system becomes apparent with 3D pole-zero plots.

Single pole system

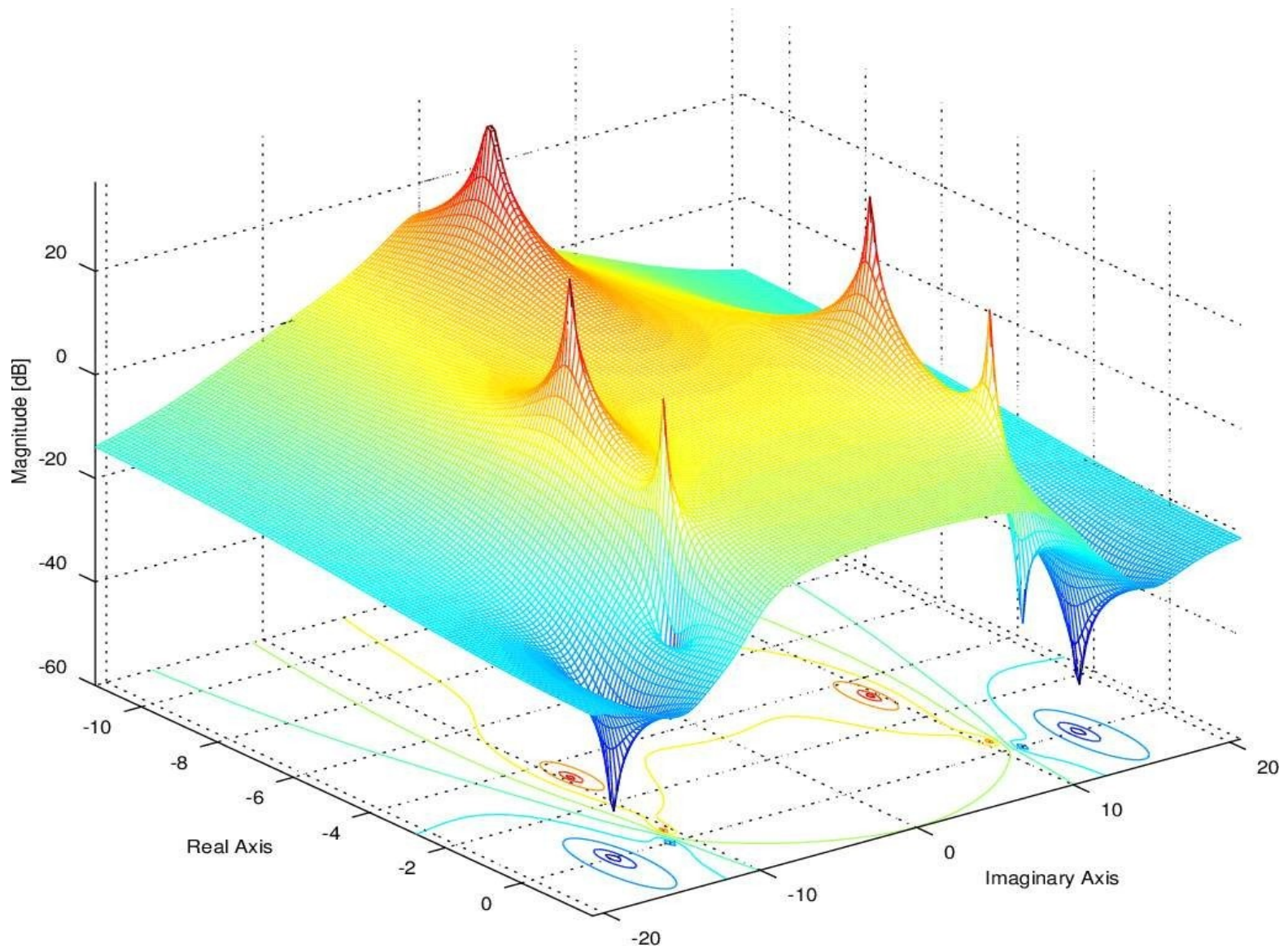


Poles and Zeros

system has 1 'zero' and 2 'poles'.



Poles and Zeros



Transfer Functions

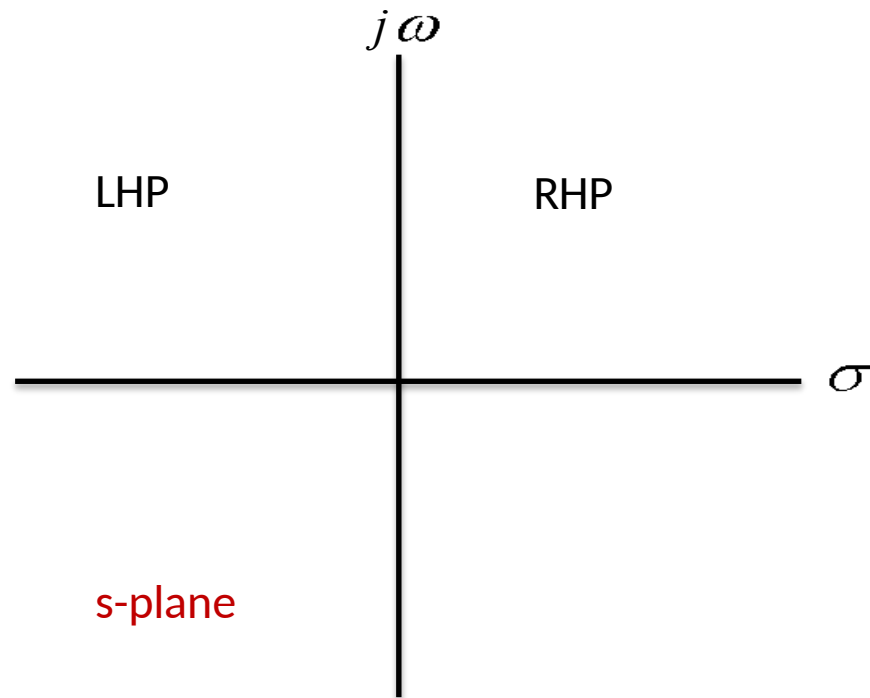
Transfer functions help us to check

- The stability of the system
- Time domain and frequency domain characteristics of the system
- Response of the system for any given input

Stability

The poles and zeros of the system are plotted in **s-plane** to check the stability of the system.

$$s = \sigma + j\omega$$

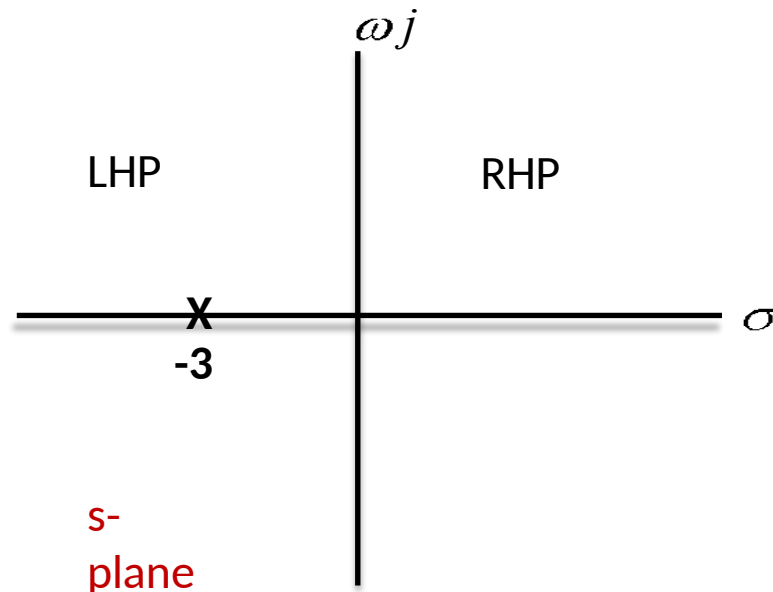


- If all the poles of the system lie in left half plane the system is said to be **Stable**
- If any of the poles lie in right half plane the system is said to be **Unstable**
- If pole(s) lie on imaginary axis the system is said to be **Marginally stable**

Simple Example

- For example
$$G(s) = \frac{C}{As + B}, \quad \text{if } A = 1, B = 3 \text{ and } C = 10$$
- Then the only pole of the system lies at

$$pole = -3$$



Tutorial 2: Have a go

- Consider the following transfer functions.
 - Determine
 - Whether the transfer function is proper or improper
 - Poles of the system
 - zeros of the system
 - Order of the system

$$\text{i)} \quad G(s) = \frac{s + 3}{s(s + 2)}$$

$$\text{ii)} \quad G(s) = \frac{s}{(s + 2)(s + 3)}$$

$$\text{iii)} \quad G(s) = \frac{(s + 3)^2}{s(s - 10)}$$

$$\text{iv)} \quad G(s) = \frac{s(s + 1)}{s^2 + s + 4}$$

Time Domain Analysis

In time-domain analysis the response of a dynamic system to an input is expressed as a function of time.

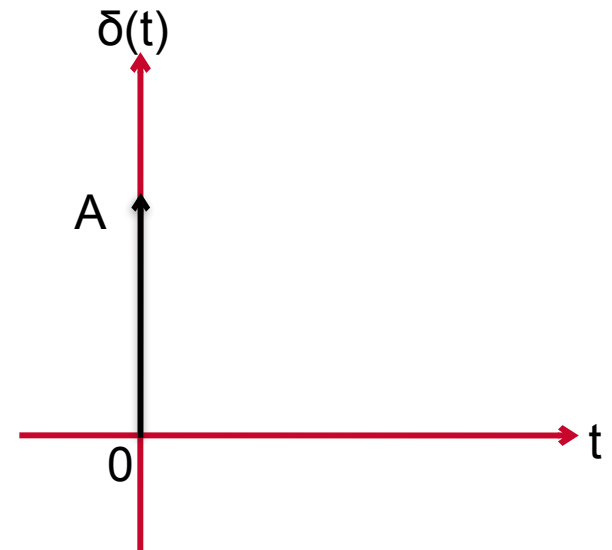
- It is possible to compute the time response of a system if the nature of input and the mathematical model of the system are known
- The characteristics of common actual input signals are a sudden shock, a sudden change, a constant velocity, and constant acceleration
- The dynamic behavior of a system is therefore judged and compared under application of standard test signals – an impulse, a step, a constant velocity, and constant acceleration
- In some situations a standard signal of great importance is a sinusoidal signal

Standard Test Signals

- The Impulse signal
 - The impulse signal imitate the sudden shock characteristic of actual input signal.
 - If $A=1$, the impulse signal is called unit impulse signal.

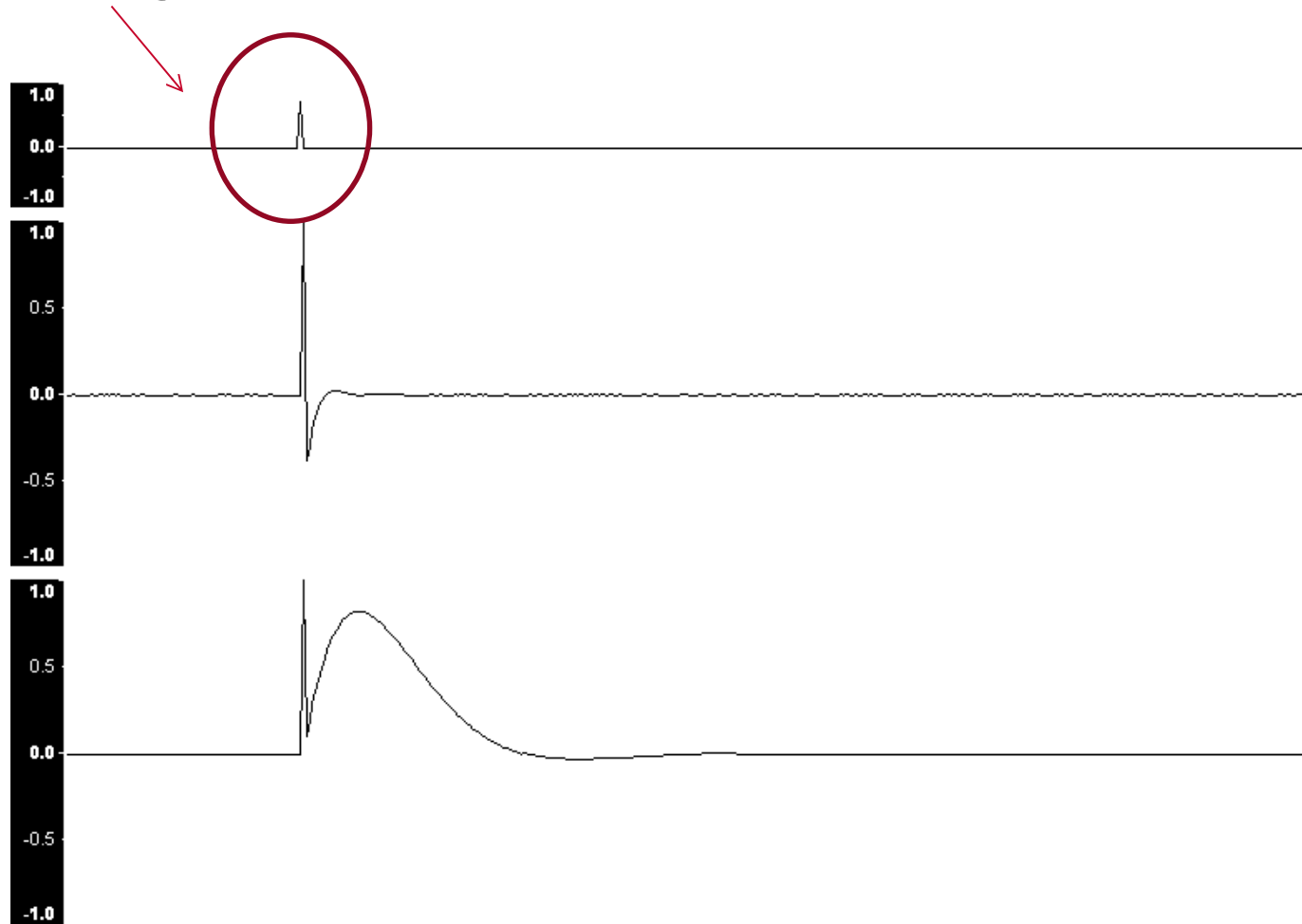
$$\delta(t) = \begin{cases} A & t = 0 \\ 0 & t \neq 0 \end{cases}$$

- If $A=1$, the impulse signal is called unit impulse signal.



Standard Test Signals

Impulse signal

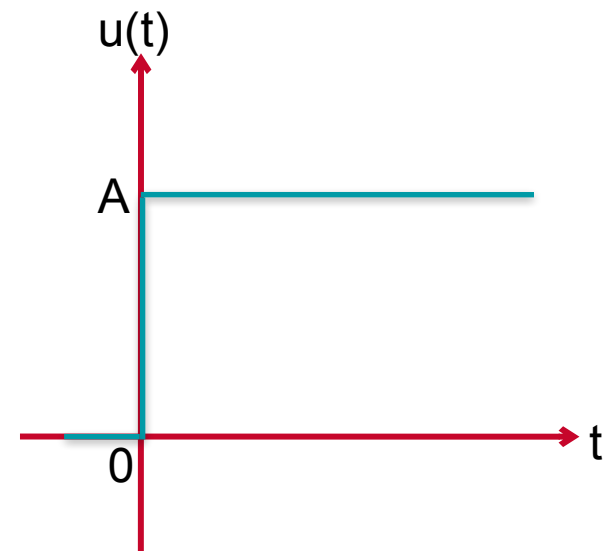


Source: English Wikipedia. [lain](#). Original image: [\[1\]](#)

Standard Test Signals

- Step signal
 - The step signal imitate the sudden change characteristic of actual input signal.

$$u(t) = \begin{cases} A & t \geq 0 \\ 0 & t < 0 \end{cases}$$

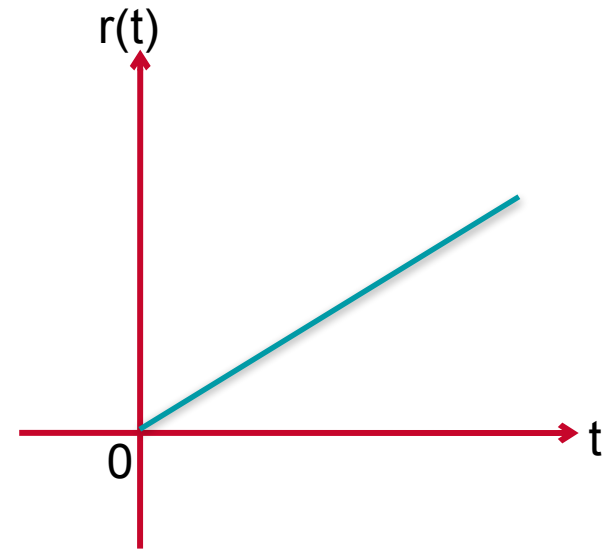


- If $A=1$, the step signal is called unit step signal

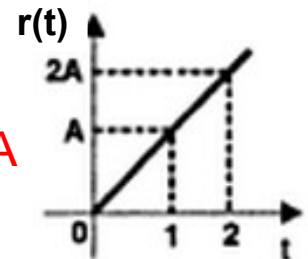
Standard Test Signals

- Ramp signal
 - The ramp signal imitate the constant velocity characteristic of actual input signal.

$$r(t) = \begin{cases} At & t \geq 0 \\ 0 & t < 0 \end{cases}$$

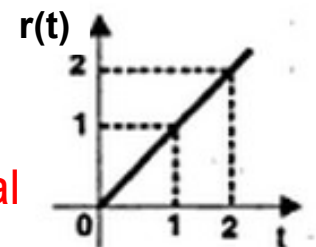


ramp signal with slope A



- If $A=1$, the ramp signal is called unit ramp signal

unit ramp signal

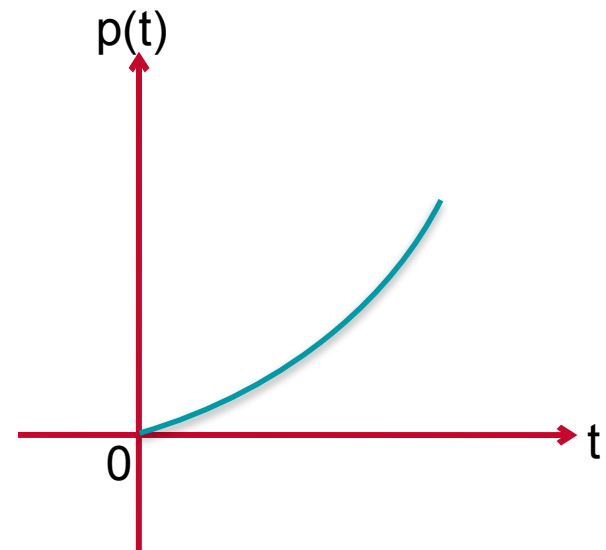


Standard Test Signals

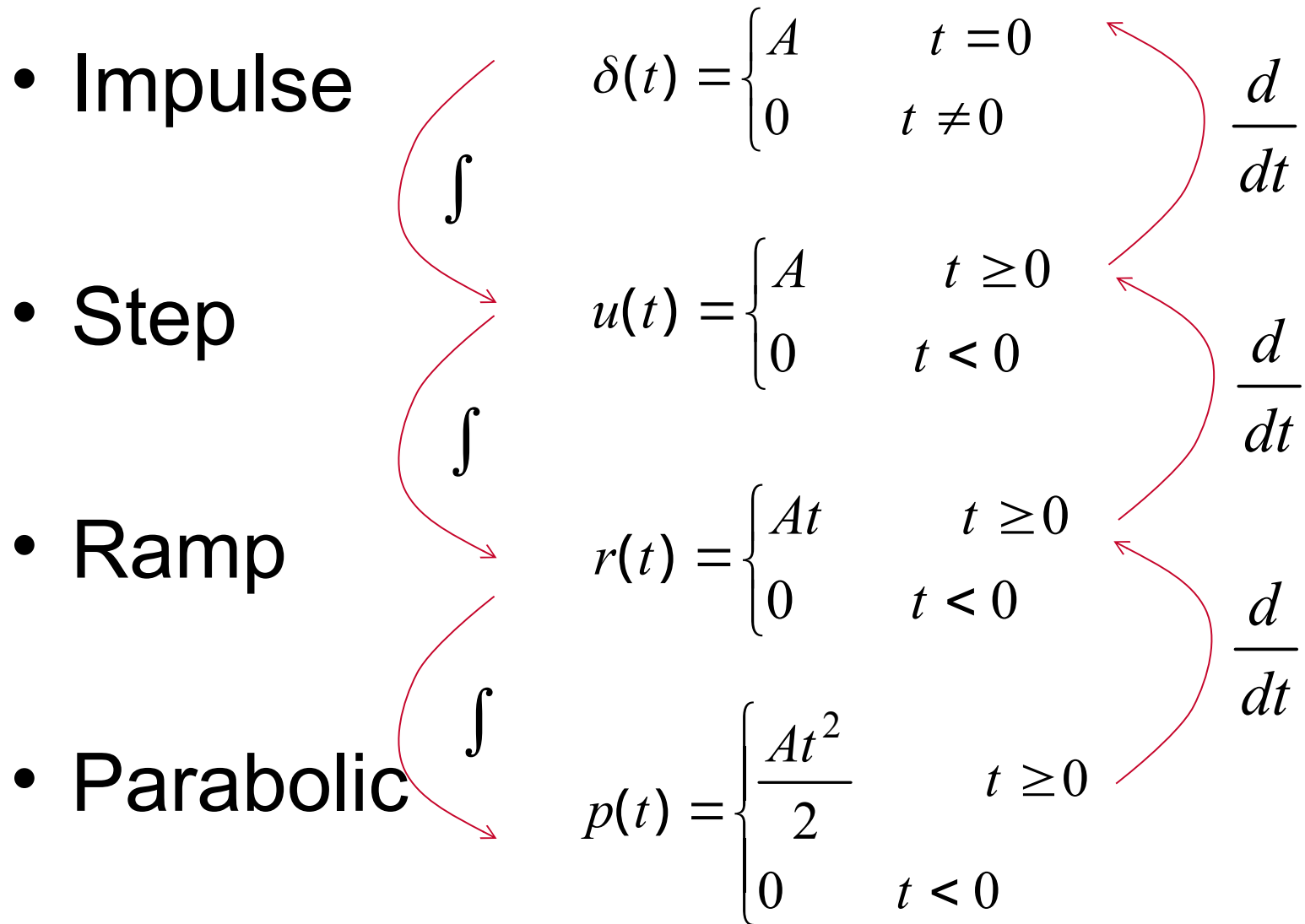
- Parabolic signal
 - The parabolic signal imitate the constant acceleration characteristic of actual input signal.

$$p(t) = \begin{cases} \frac{At^2}{2} & t \geq 0 \\ 0 & t < 0 \end{cases}$$

- If $A=1$, the parabolic signal is called unit parabolic signal.



Relation Between Standard Test Signals



Laplace Transform of Test Signals

- Impulse

$$\delta(t) = \begin{cases} A & t = 0 \\ 0 & t \neq 0 \end{cases}$$

$$L\{\delta(t)\} = \delta(s) = A$$

- Step

$$u(t) = \begin{cases} A & t \geq 0 \\ 0 & t < 0 \end{cases}$$

$$L\{u(t)\} = U(s) = \frac{A}{s}$$

Laplace Transform of Test Signals

- Ramp

$$r(t) = \begin{cases} At & t \geq 0 \\ 0 & t < 0 \end{cases}$$

$$L\{r(t)\} = R(s) = \frac{A}{s^2}$$

- Parabolic

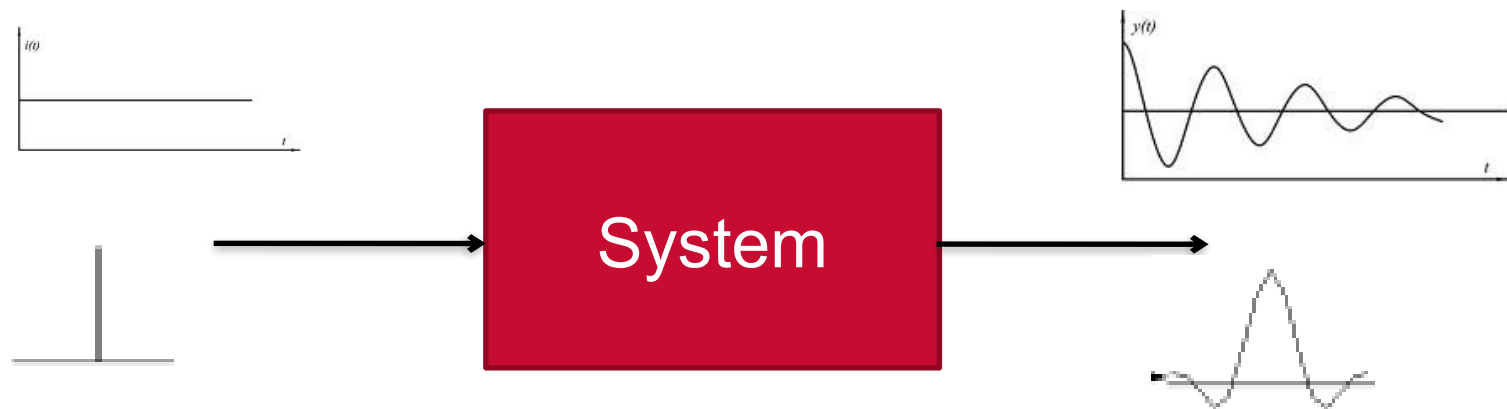
$$p(t) = \begin{cases} \frac{At^2}{2} & t \geq 0 \\ 0 & t < 0 \end{cases}$$

$$L\{p(t)\} = P(s) = \frac{2A}{s^3}$$

Time Response of Control Systems

The time response of a dynamic system is a response to an input expressed as a function of time.

- The time response of any system has two components
 - Transient response
 - Steady-state response.

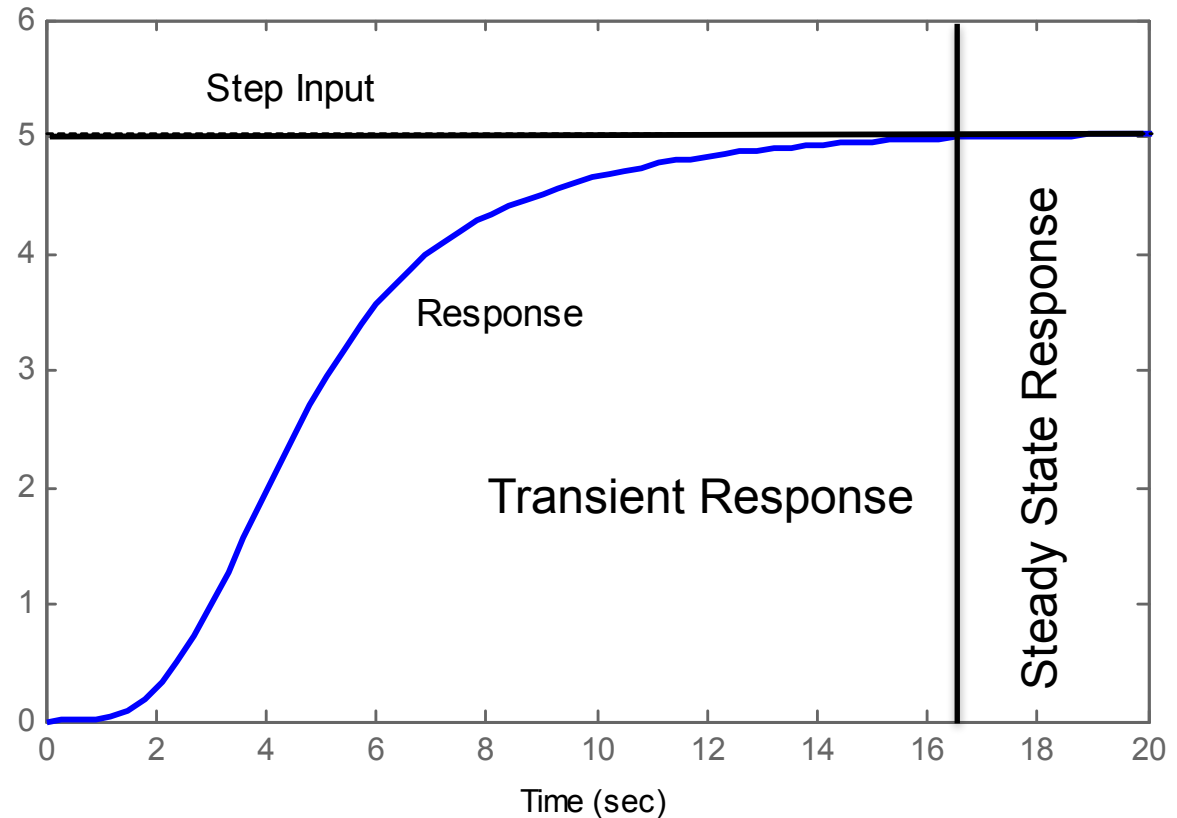


Time Response of Control Systems

When the response of the system is changed from rest or equilibrium it takes some time to settle down.

- The transient response is the response of a system from rest or equilibrium to steady state.

- The response of the system after the transient response is called steady response.



Time Response of Control Systems

The transient response depends upon the system poles only and not on the type of input

- It is therefore sufficient to analyze the transient response using a step input.
- The steady-state response depends on system dynamics and the input quantity.
- It is then examined using different test signals.

Time Responses of First Order systems

The first order system has the following generic format with only one pole.

$$\frac{C(s)}{R(s)} = \frac{K}{Ts + 1}$$

Where **K** is the D.C gain and **T** is the time constant of the system.

- The time constant is a measure of how quickly a 1st order system responds to a unit step input.
- The D.C Gain of the system is ratio between the input signal and the steady state value of output.

For the first order systems given below

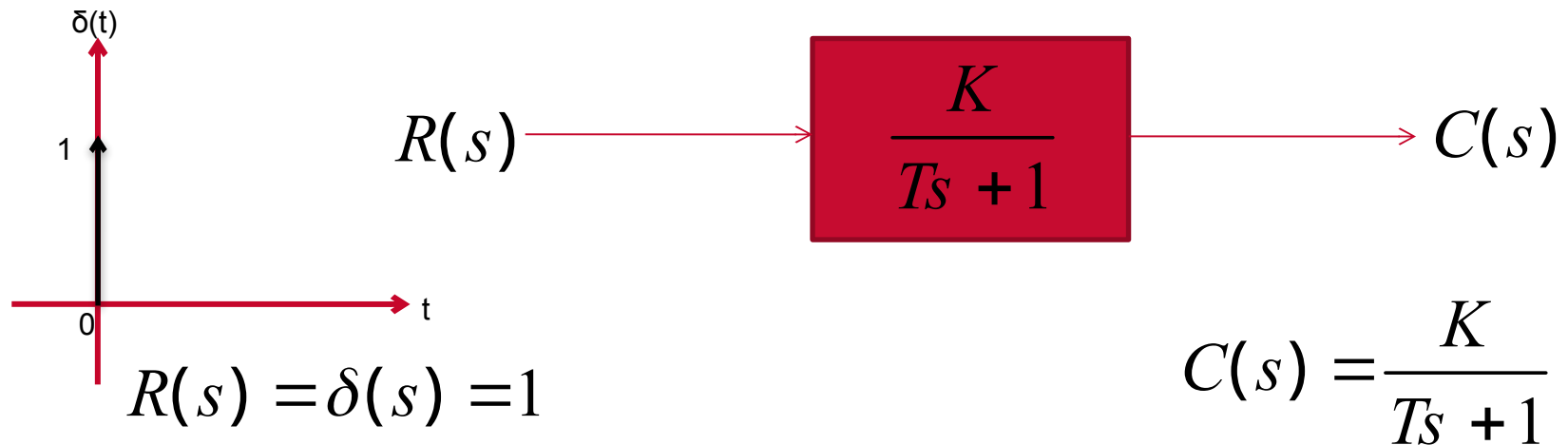
$$G(s) = \frac{10}{3s + 1}$$

D.C gain is **10**
time constant is **3** seconds

$$G(s) = \frac{3}{s + 5} = \frac{3/5}{1/5s + 1}$$

D.C gain is **3/5**
time constant is **1/5** seconds.

Impulse Response of First Order System



$$C(s) = \frac{K/T}{s + 1/T}$$

- In order to represent the response of the system in time domain we need to compute inverse Laplace transform of the above equation.

$$L^{-1}\left(\frac{A}{s + a}\right) = Ae^{-at}$$

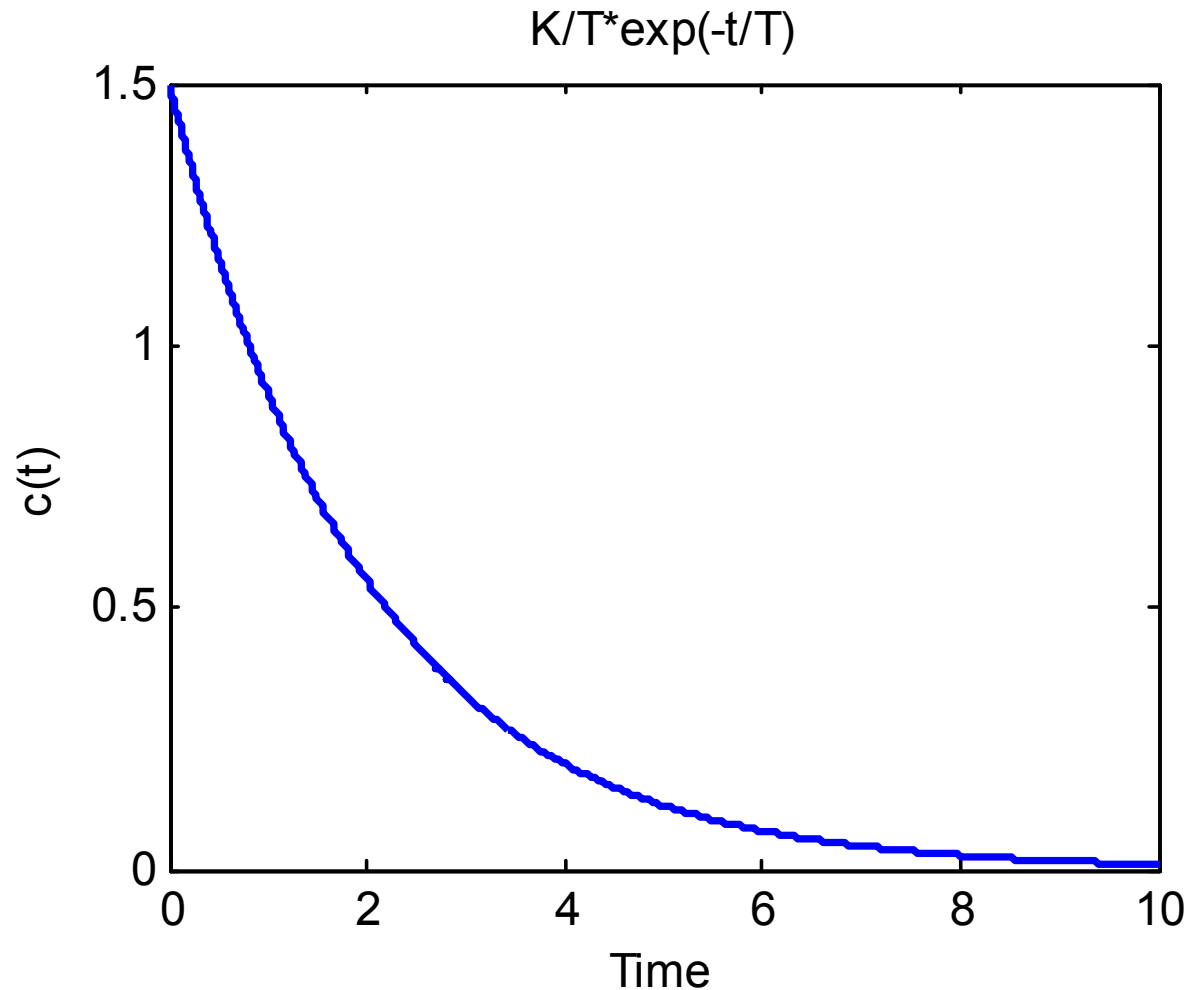
$$c(t) = \frac{K}{T} e^{-t/T}$$

Impulse Response of First Order System

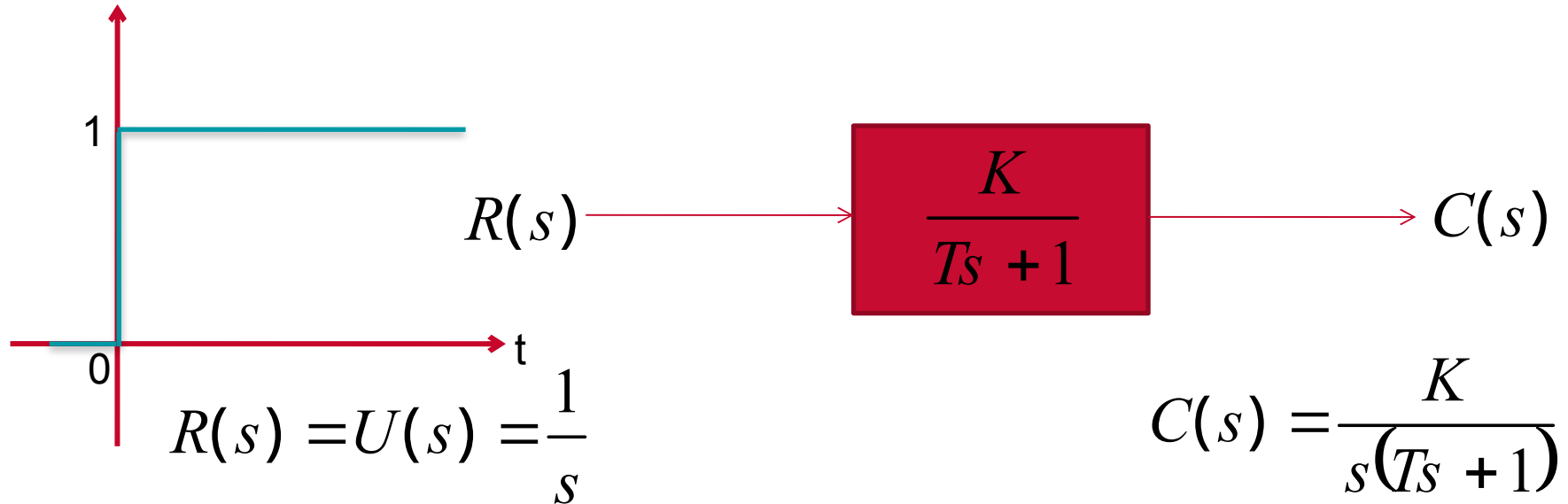
Example.

- If $K=3$ and $T=2s$ then

$$c(t) = \frac{K}{T} e^{-t/T}$$



Step Response of First Order System



Forced Response

$$C(s) = \frac{K}{s} + \frac{KT}{Ts + 1}$$

Natural Response

$$C(s) = K \left(\frac{1}{s} - \frac{T}{Ts + 1} \right)$$

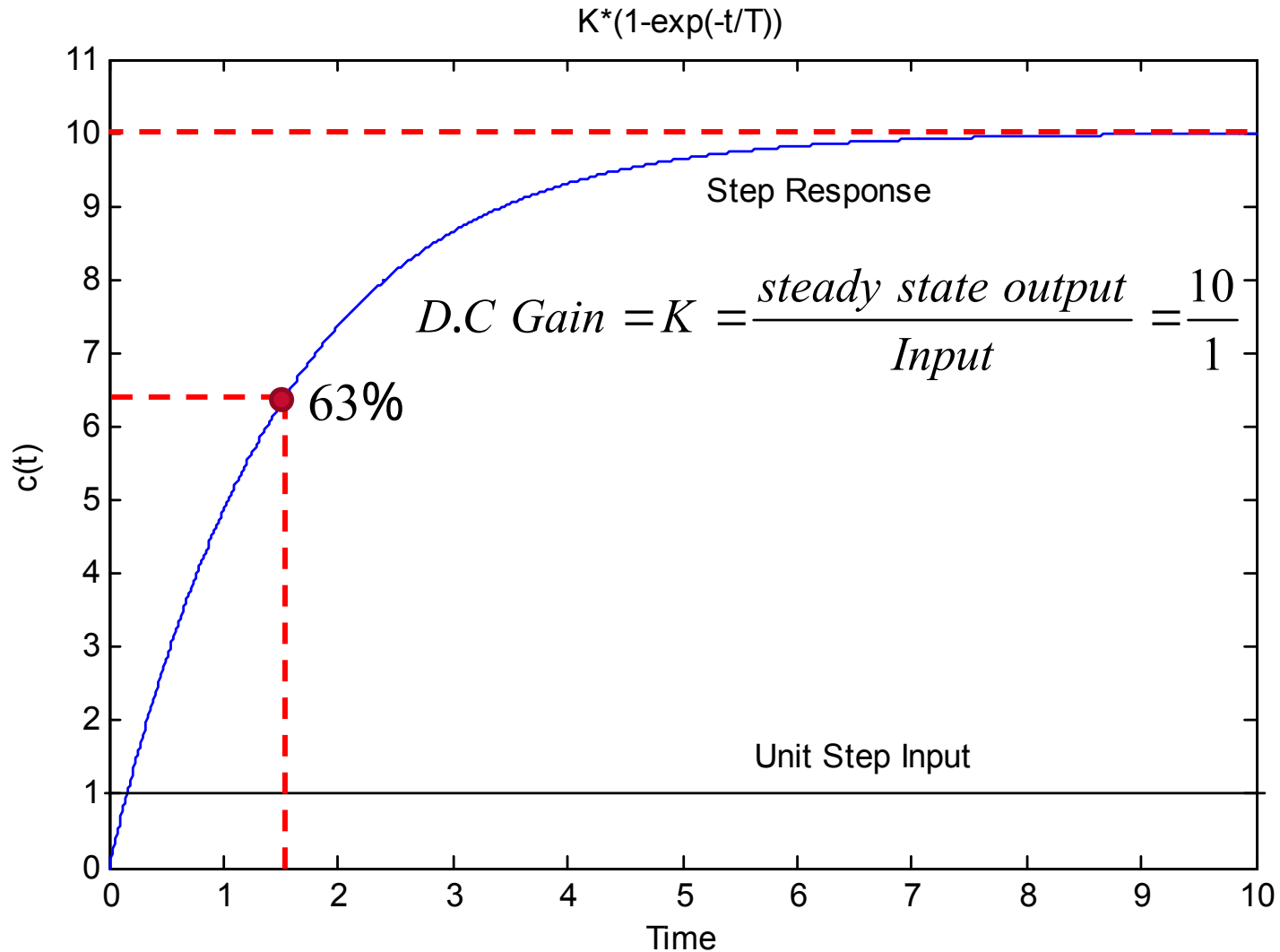
- Taking inverse Laplace transform of the above equation.

$$c(t) = K(u(t) - e^{-t/T}) \rightarrow \text{if } u(t)=1 \rightarrow c(t) = K(1 - e^{-t/T})$$

Step Response of First Order System

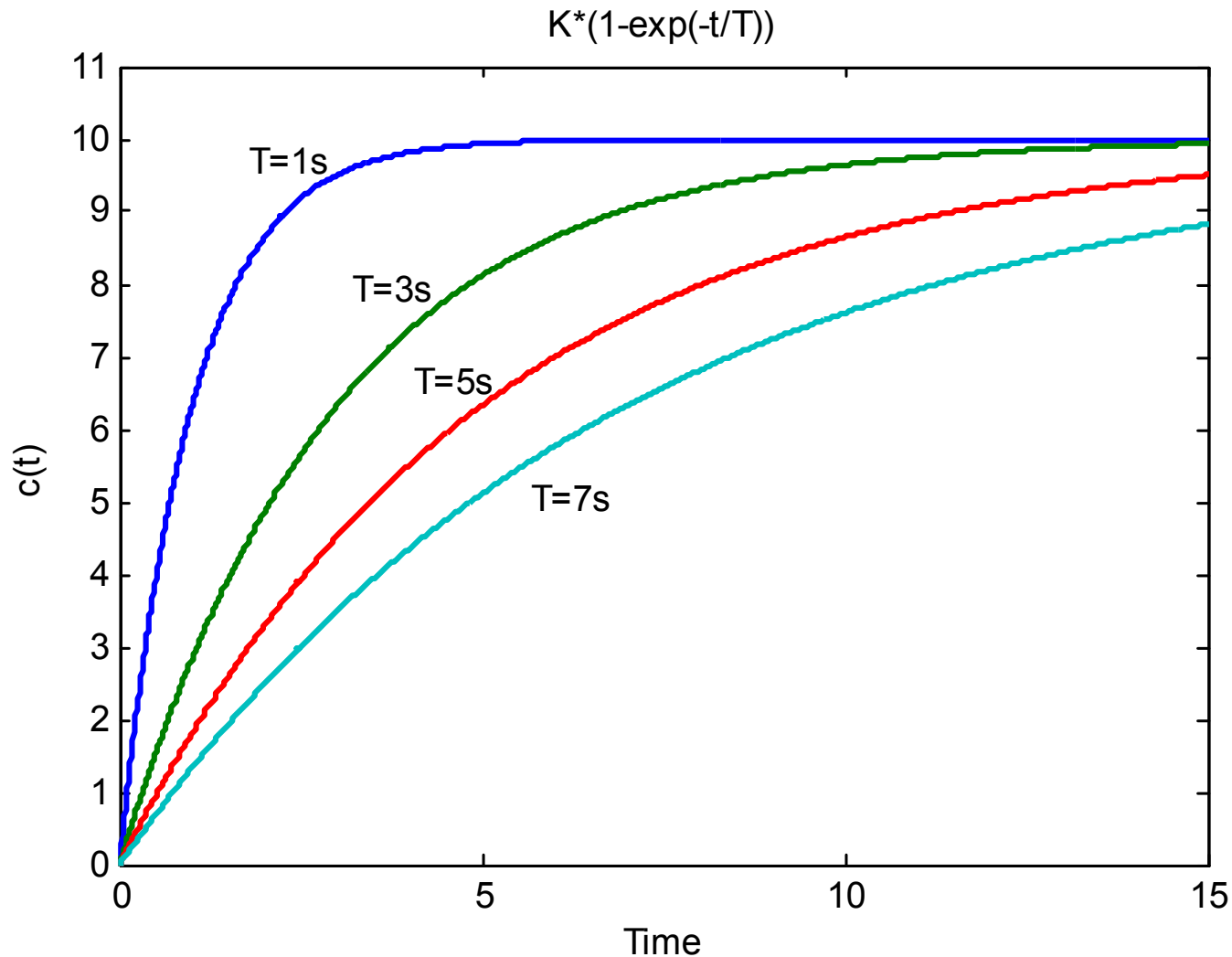
Example.

- If $K=10$ and $T=1.5s$ then using $c(t) = K(1 - e^{-t/T})$



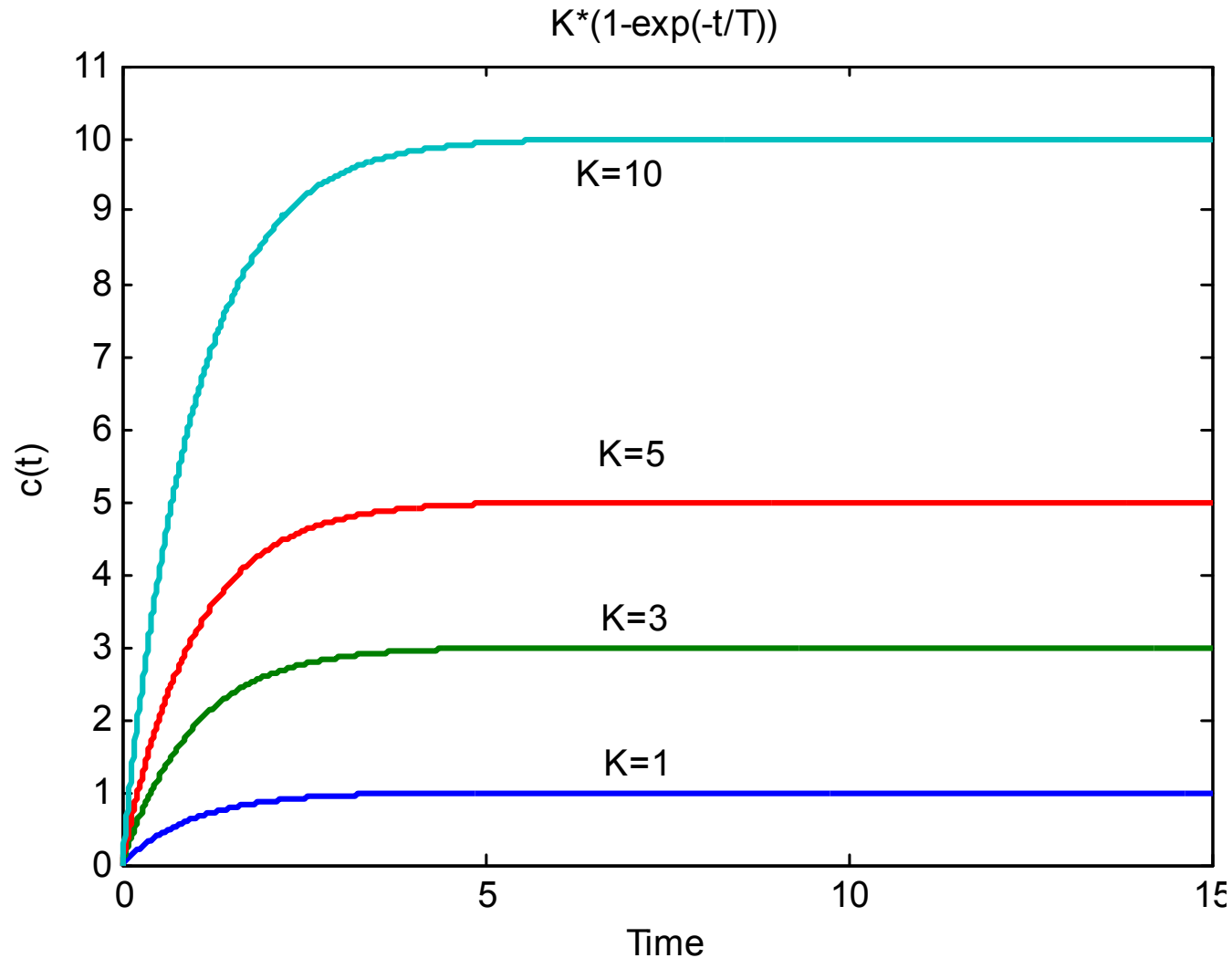
Step Response of First Order System

- If $K=10$ and $T=1, 3, 5, 7$ using $c(t) = K(1 - e^{-t/T})$

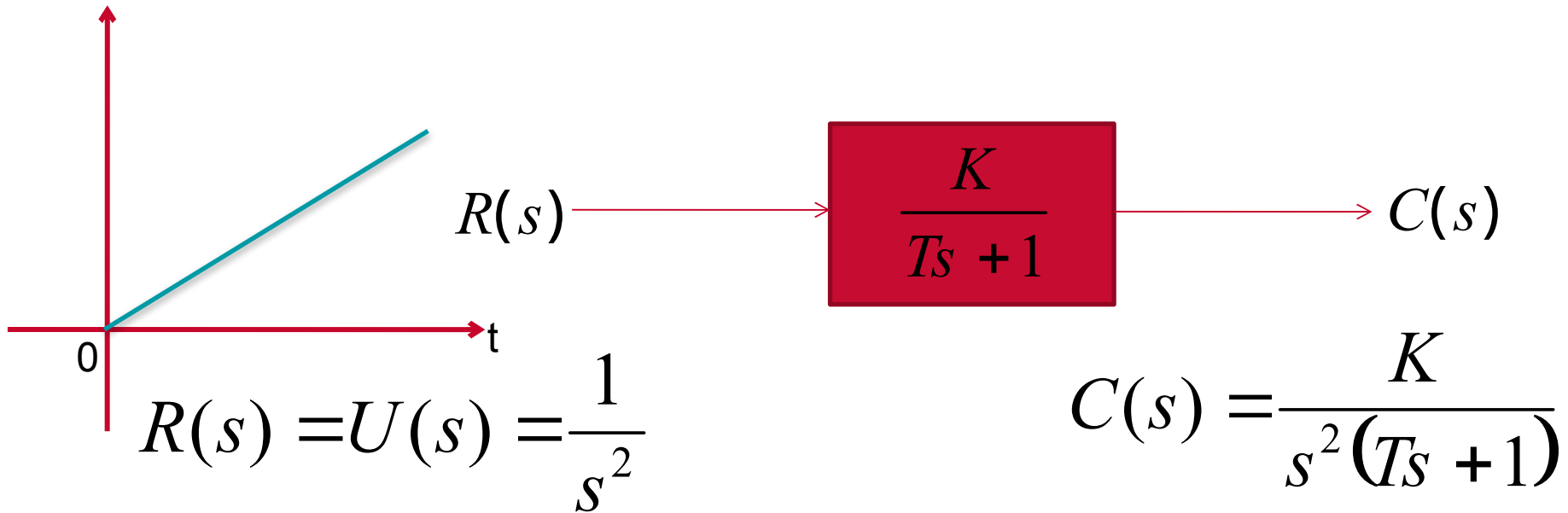


Impulse Response of First Order System

- If $T=1$ and $K=1, 3, 5, 7$ using $c(t) = K(1 - e^{-t/T})$



Ramp Response of First Order System



$$C(s) = K \left(\frac{1}{s^2} - \frac{1}{s} - \frac{T}{Ts + 1} \right)$$

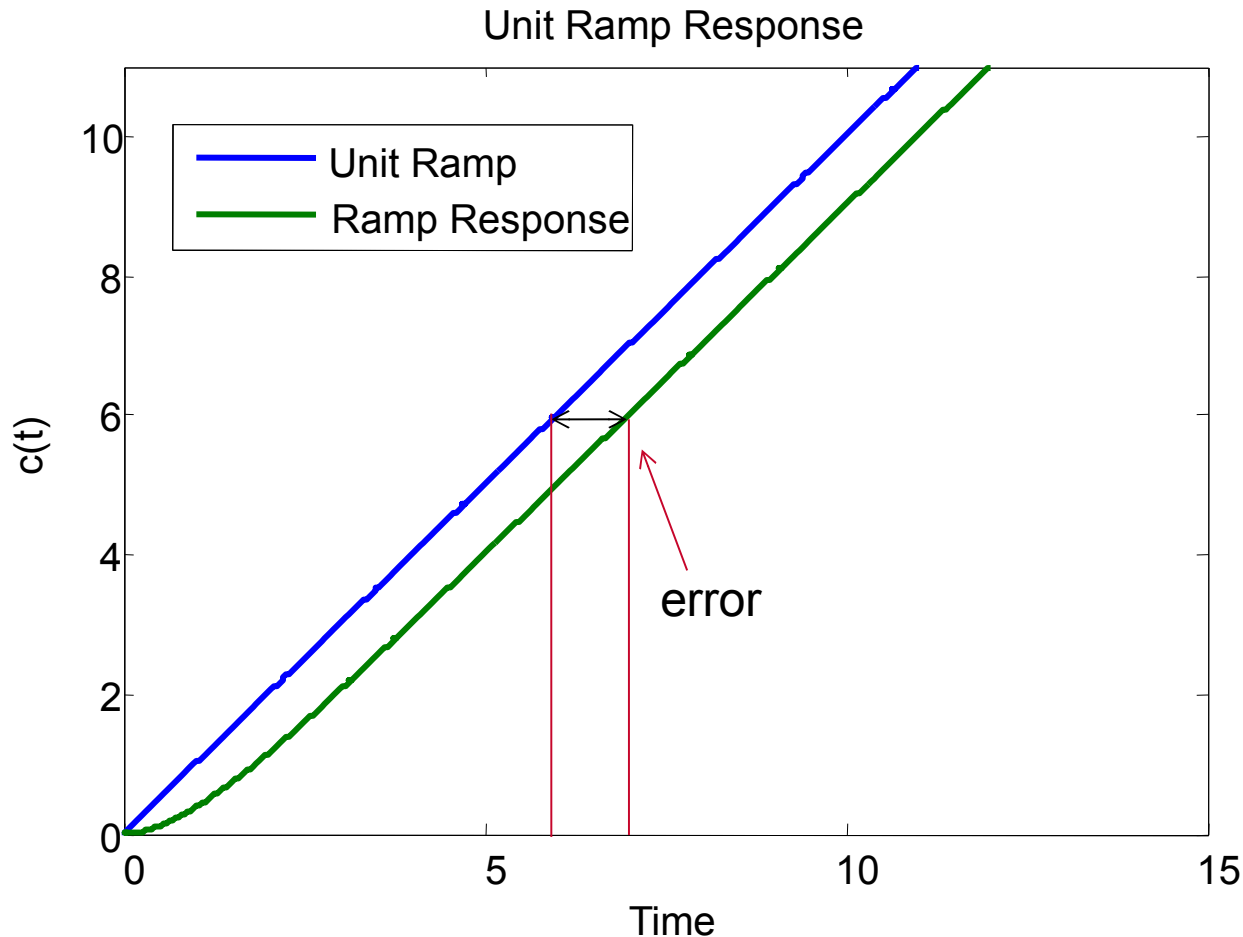
- Taking inverse Laplace transform of the above equation.

$$c(t) = K(t - T + Te^{-t/T})$$

Ramp Response of First Order System

Example.

- If $K=1$ and $T=1$ then using $c(t) = K(t - T + Te^{-t/T})$

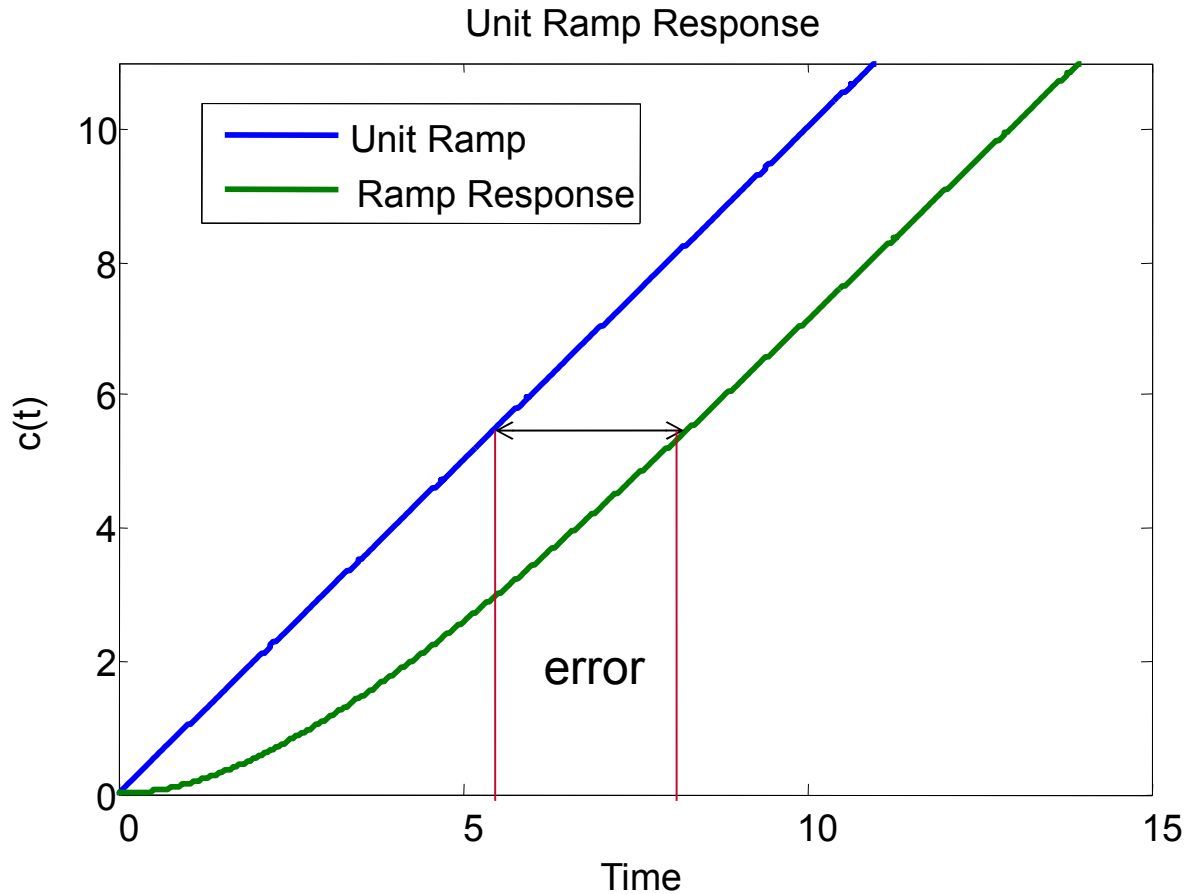


$$e(t) = r(t) - c(t) = T(1 - e^{-t/T})$$

Ramp Response of First Order System

Example.

- If $K=1$ and $T=3$ then using $c(t) = K(t - T + Te^{-t/T})$



$$e(t) = r(t) - c(t) = T(1 - e^{-t/T})$$

Parabolic Response of First Order System

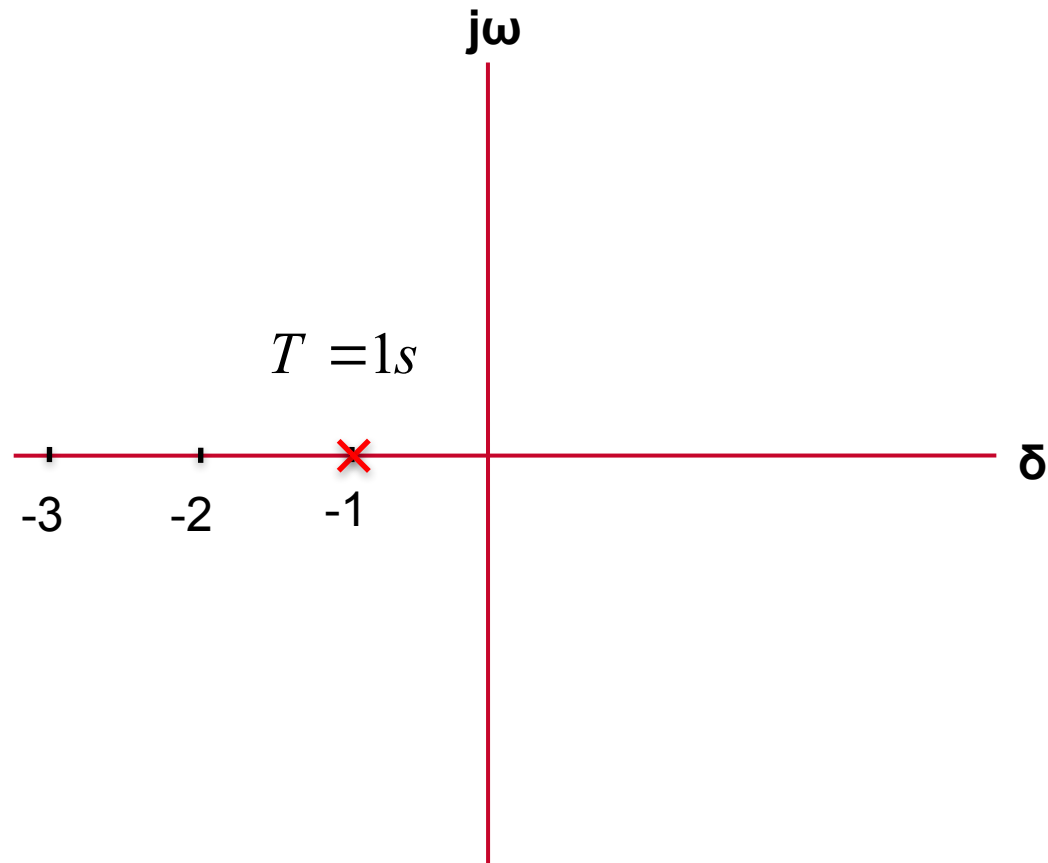
The parabolic response is much the same as the ramp response for a first order system as it just produce the curve with a delay based on the time constant.

- The ramp and parabolic inputs are known as unbounded as they keep changing and do not lead to a steady state response.
- So it is the step response that is most useful to us in this case.
 - This can be useful if the system is closed, and the component parts are not easily identifiable.
 - The system's step response can lead to a representation even though the inner construction is not known
 - With a step input, we can measure the time constant and the steady-state value, from which the transfer function can be calculated

PZ-map and Step Response

$$\frac{C(s)}{R(s)} = \frac{K}{Ts + 1}$$

$$\frac{C(s)}{R(s)} = \frac{10}{s + 1}$$

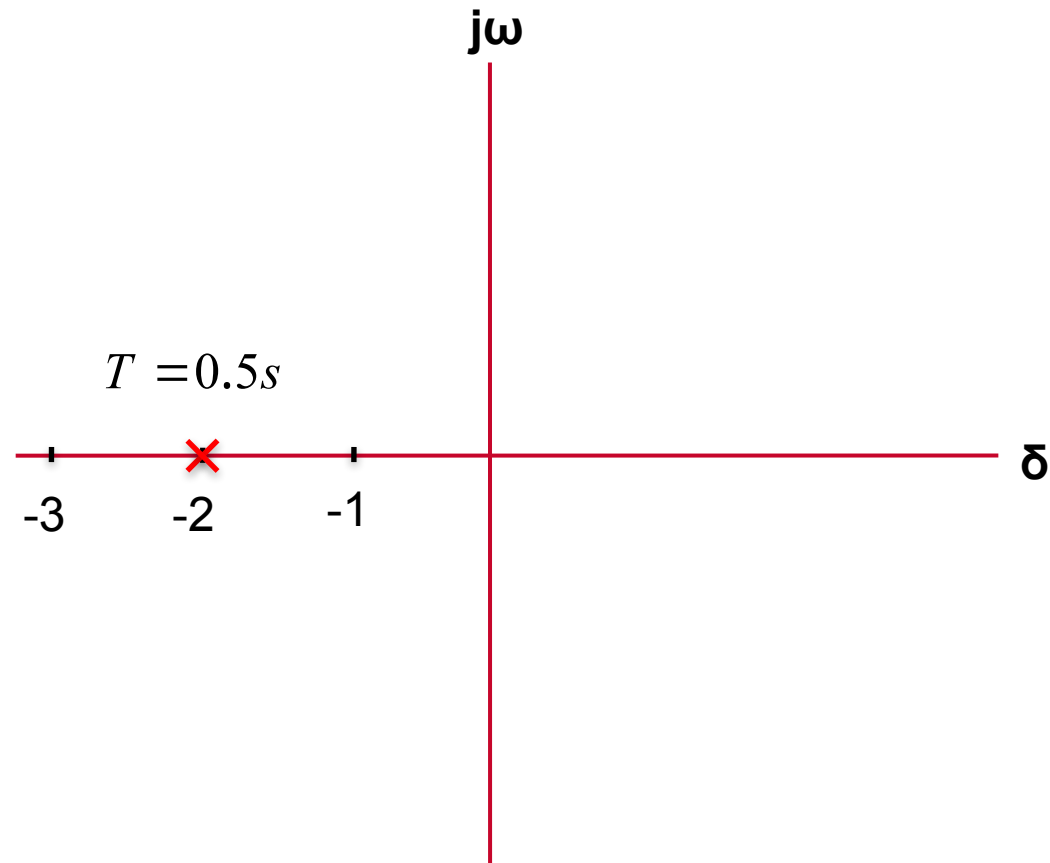


PZ-map and Step Response

$$\frac{C(s)}{R(s)} = \frac{K}{Ts + 1}$$

$$\frac{C(s)}{R(s)} = \frac{10}{s + 2}$$

$$\frac{C(s)}{R(s)} = \frac{5}{0.5s + 1}$$

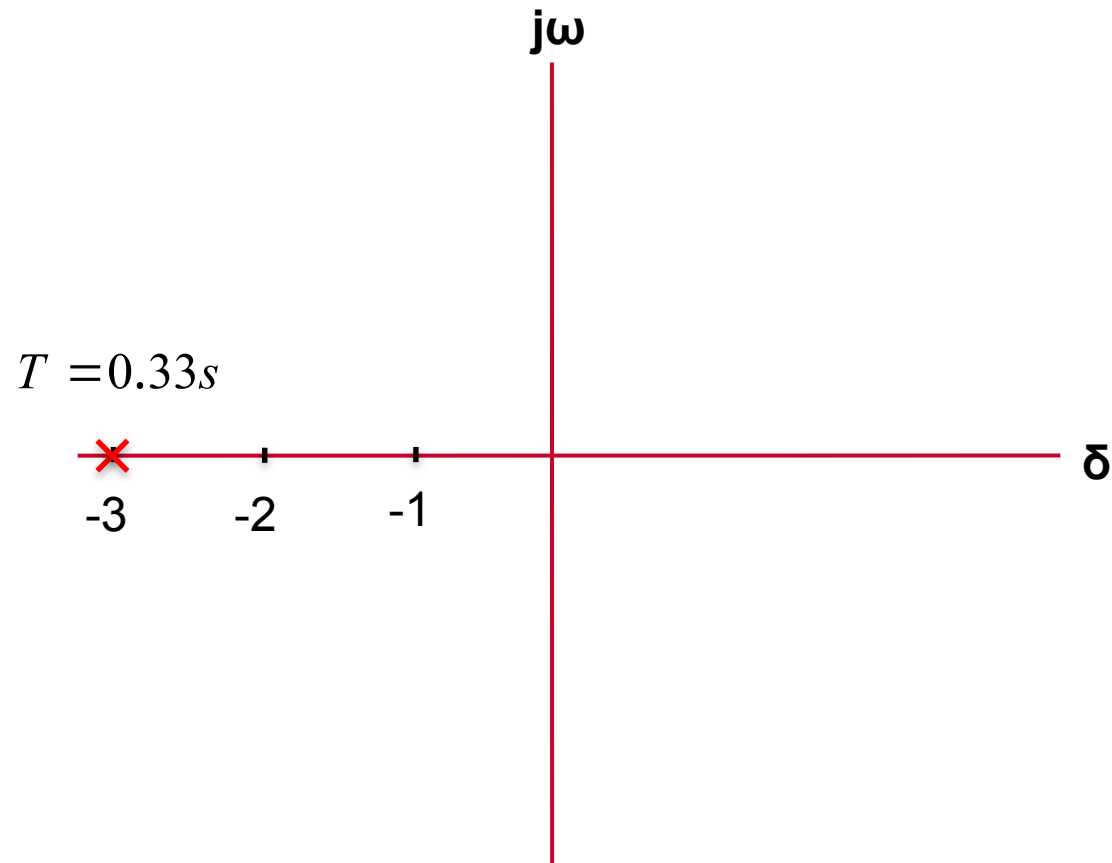


PZ-map and Step Response

$$\frac{C(s)}{R(s)} = \frac{K}{Ts + 1}$$

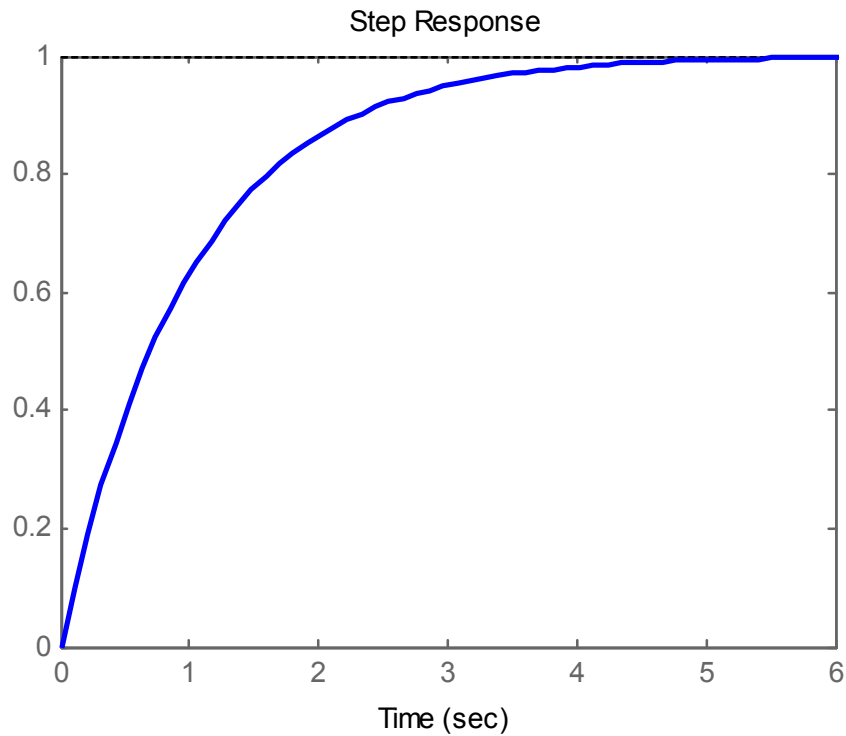
$$\frac{C(s)}{R(s)} = \frac{10}{s + 3}$$

$$\frac{C(s)}{R(s)} = \frac{3.3}{0.33s + 1}$$



Comparison

$$\frac{C(s)}{R(s)} = \frac{1}{s + 1}$$



$$\frac{C(s)}{R(s)} = \frac{1}{s + 10}$$

